

Examiner reports

Q1.

Question (a) was completed successfully by most students. The most common error was a sign mistake.

Well over half of the students scored at least two marks on part (b). The best students realised they had to do very little to answer this question and, provided they explained their arguments rigorously, they achieved full marks. Most students completed more calculations than necessary (there was no need to find the lengths of all sides), but failed to complete their arguments in sufficient detail. For example, they may have explained why the shape was a parallelogram, but not explained why it was not a rhombus.

Q4.

Again this question was done well by many students although some made minor errors. In part (c) a few students found the acceleration (as $1.4\mathbf{i} + 1.6\mathbf{j}$) and did not calculate the magnitude. Also, in part (d) some students found the angle as 41° rather than 49° . However many students gained full marks on this question.

Q5.

Part (a) was the most accessible part of this question. While there were a good number of correct responses, there were also a significant number of students who had problems with the initial position. Often this was initially placed on the wrong side of the equals sign so that the students ended up with $-(500\mathbf{i} + 200)$ in their expressions for the position vector.

Part (b) was much more challenging for the students. Those who were able to form two quadratic equations from the components of the position vector were usually able to go on and solve these to find the correct time. Incorrect approaches included trying to work with the magnitude of the position vector or not using the position vector of the rock.

Part (c) was also problematic for some students. Many ended up with $\frac{300\mathbf{i} + 600\mathbf{j}}{\text{time}}$ instead of $\frac{-300\mathbf{i} - 600\mathbf{j}}{\text{time}}$.

The explanations in part (d) were generally inadequate, with answers like velocity has direction and speed does not, which did not go far enough to justify the award of the mark.

Q6.

This question was also answered well. In part (a), a few candidates did not divide $\mathbf{F}$ by 50 accurately. In part (b), a number of candidates did not find their ' $\mathbf{c}$ ' correctly, simply replacing their ' $\mathbf{c}$ ' with $7\mathbf{i} - 4\mathbf{j}$. Others, on evaluating their ' $\mathbf{c}$ ', took $-e^{-2t}$ when $t = 0$ to be $+1$ rather than -1 .

Q7.

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available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).

Part (c) was altogether more challenging and many candidates worked with position vectors rather than velocity vectors. This clearly was the wrong approach, but some expended a lot of effort solving the quadratic equations that they produced. Some candidates came up with the correct velocity, but without showing how they had obtained their answer. It is important that working is justified. Sometimes a trial and improvement method led candidates to the correct answer.

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Q9.

Parts (a) and (b) were generally done well, with the candidates gaining many marks. A few candidates could not state the direction in part (b)(ii) and gave answers such as north-east or south-west. Part (c) was found to be more demanding. Candidates who did not identify the $\mathbf{i}$ and $\mathbf{j}$ components of the velocity made very little progress. Some made some initial good progress, but made arithmetic or algebraic errors in their solutions.

Some forgot to square the speed and tried to work from equations such as $(4 - 0.4t)^2 + (0.5 + 0.2t)^2 = 5$. Some candidates used a trial and improvement approach and realised that the velocity must be $-3\mathbf{i} + 4\mathbf{j}$, which produced a speed of 5.

Q10.

Most candidates answered this question well. In part (a), a small number of candidates forgot to add, and find, the $+c$ term and thus ignored the initial position vector; a few candidates tried to use equations for constant acceleration. In part (c), some candidates made errors when finding the force by finding $m\mathbf{a}$, which was $2(6t\mathbf{i} - 8\mathbf{j})$, to be $12t\mathbf{i} - 8\mathbf{j}$.

Q11.

In part (a), many candidates produced correct answers, but some stopped when they had obtained the velocity and did not go on to find the speed as requested. Part (b) was generally done well, but a few candidates did not give their final answer to the nearest degree. Part (c) was very much more demanding. A significant number of candidates mixed scalar and vector quantities in an attempt to answer the question. Statements like

$500 = \frac{1}{2} (0.5\mathbf{i} + 0.375\mathbf{j})^2$ were quite common and there were some similar ones that also

included an initial velocity. The successful candidates used a variety of approaches, which can be seen in the mark scheme. The key difference between those who were successful and those who were not was the ability to connect the scalar distance of 500 to the position vector in a meaningful way.

Q12.

Most candidates showed that they knew the techniques involved in answering this question. Candidates generally answered part (a) correctly. Part (b)(i) was answered well, but in part (b)(ii) some candidates forgot to find the magnitude of $\mathbf{F}$, giving instead just the vector $\mathbf{F}$, $-40\mathbf{i} + 30\mathbf{j}$. Most candidates answered part (c) correctly. In part (d) the majority of candidates integrated to find the position vector, but a common error was to forget the $+\mathbf{c}$ term and thus to ignore the initial position vector. Some found the $+\mathbf{c}$ incorrectly, assuming that $\mathbf{c}$ was the initial position vector.

Q13.

Most candidates answered parts (a), (b) and (c) well, although some did get an answer of $-4.5\mathbf{j}$ for part (b).

In part (c), the candidates who clearly stated and used a constant acceleration equation usually made a good start and completed the question, although there were a few arithmetic errors.

In part (e) many candidates did not attempt to find the magnitude of the force.

Part (f) was generally found more difficult with only a few good attempts. Several candidates did not attempt this part of the question.

Q14.

Parts (a) and (b) of this question were done very well and many candidates gained all four marks.

Part (c) proved to be more challenging. In part (c)(i), many candidates did not show how they started their working. An equation based on $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ was expected, but not always seen. Some candidates obtained the printed answer, but did not gain full marks as their starting point was not clear. Having both the acceleration and the initial velocity given in the question enabled many candidates to write down a correct expression for the velocity of the particle. Part (c)(iii) was found to be very challenging. If candidates were able to write down an expression for the speed of the particle they would often go on to find the times correctly. It was this vital first step that many candidates were unable to take. A few candidates made errors solving the quadratic equation once they had obtained it correctly.

Q15.

The majority of the candidates were able to find the initial velocity as requested in part (a), but most found the rest of the question difficult. A reasonable number of the candidates were able to find the acceleration, some by calculating two velocities and using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, but others clearly could not find a strategy to find the acceleration. Part (c) was found to be more demanding. Some candidates did use the components correctly, but quite a number of candidates did not recognise the approach that was required.

Q16.

While some candidates did well with this question, others made some fundamental errors. In part (a), some candidates found the magnitude of the acceleration and then tried to use this scalar quantity in the later parts of the question, ending up with a mix of scalar and vector quantities.

In part (b)(i), many candidates found the position vector, but did not go on to find the distance as requested. There were quite a number of correct responses to part (b)(ii), but quite often this was not followed by a correct approach to the final part of the question.

Q17.

Virtually all candidates answered part (a) correctly. The common problem in part (b) was

in the integration of $400 \cos \frac{\pi}{2} t$. Apart from this, answers to part (b) were good, and most correctly found $t = 2$ in part (c). The majority of candidates also found correctly that the speed of the particle was 3, with only a few finishing with the velocity of the particle being -3 or $-3i$.

Q18.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

Part (b) was much more demanding and relatively few correct solutions were seen. Those candidates who identified the required method were able to do well, but the majority made no attempt or produced working which could not lead to the required answer. Similarly part (c) was found to be quite difficult, but some candidates did use the answer from part (b) to put together correct solutions for part (c) when they had been unable to do part (b).

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Q19.

Parts (a) and (b) were generally done well by the candidates. A few candidates subtracted the vectors in part (a) instead of adding them. Also some tried to find the magnitude of each vector in part (a) and then lost several marks because they had very confused working.

In part (c) there were many attempts which involved the division of vectors. For example

working such as: $m = \frac{9i + 12j}{1.5i + 2j} = 6i + 6j$. Candidates should be advised to consider the components involved and not to attempt to divide vectors.

There were a number of errors that appeared in part (d). One was to include an initial velocity, which was equal to the resultant force on the particle. A few candidates used incorrect formulae. In the final part of the question a large number of candidates gave the vector $3i + 4j$ as their final answer and did not find the magnitude in order to get the distance as requested.

Q20.

Most of this question was answered well. In part (a), the only difficulty was found in

differentiating $2e^{\frac{1}{2}t}$. In part (b), many found the velocity of the particle to be $(e^{\frac{3}{2}} - 8)\mathbf{i}$ [or $-3.52\mathbf{i}$], but did not find the speed of $+3.52 \text{ m s}^{-1}$. Others gave the speed as 3.5 m s^{-1} , which was penalised. Those candidates who were successful in part (a) usually completed parts (c) and (d) correctly.

Q21.

This question was demanding for the candidates, although part (a) caused very few difficulties for the majority, and some gained only the three marks for this part of the question.

Part (b) was the most demanding part of the question, and only very few candidates gained the marks available. Many candidates stated that the $\mathbf{i}$ component should be zero, but very few formed an equation to find the time when this would be. Most incorrect approaches were based on using a time, often 20 seconds, and then calculating the $\mathbf{j}$ component of the velocity and stating that the $\mathbf{i}$ component was 0.

There were some good answers to part (c), but quite a few candidates failed to include the initial position of the particle.

In part (d), there were some good solutions, but some candidates made errors. These

included not subtracting the initial position. Some candidates used $\frac{\mathbf{u} + \mathbf{v}}{2}$. With this method there were two main errors: dividing by 20 rather than 2 and using an incorrect expression for either $\mathbf{u}$ or $\mathbf{v}$.

Q22.

In this question the use of vector notation by some candidates was quite poor. Part (a) was done well by a good number of candidates, but some did not show enough working to gain full marks. Typical issues were not showing that their equation was derived from $\mathbf{v} = \mathbf{u} + \mathbf{at}$ and not showing the division by 40 clearly. Some candidates did well with part (b), but there were quite a lot of arithmetic errors.

Some candidates used an incorrect initial velocity, for example $4\mathbf{i}$ instead of $5\mathbf{j}$. Part (c) proved to be too demanding for the majority of the candidates, although some were able to gain a few marks. One of the most common errors was to work with position vectors rather than with velocities. For those who did work with velocities, creating an equation to find the time when the Jet Ski was travelling south east was found to be difficult. Quite a number of the candidates who got this far had equations that lacked a minus sign.

Q23.

This question was generally answered well. The only common error was in part (b)(ii), where a number of candidates found $\mathbf{F}$ to be $36\mathbf{i} + 6\mathbf{j}$, but did not find its magnitude.

Q24.

This question was also done well by many candidates. In part (a), there were some confused responses and occasional problems with minus signs. Part (b) was also

generally done well. In part (c), a few candidates had problems. The most common of these was to omit an arrow from the diagram to show the force; the other was to use the

lengths incorrectly when calculating the angle, for example $\tan^{-1}\left(\frac{8}{2}\right)$ and $\sin^{-1}\left(\frac{8}{8.25}\right)$.

Q25.

Generally the candidates made quite good attempts at this question, but some parts caused the candidates more difficulty. A few candidates made poor use of vector notation. Part (a) was answered correctly by the vast majority of the candidates. The most common error was dealing incorrectly with the initial velocity. A few went on to simplify their answers incorrectly. Part (b) was a little more demanding, as some candidates confused the components. Part (c) did cause a little more difficulty for some candidates, with a few candidates not giving any answer at all. However, there were again many correct solutions.

In part (d)(i), although there were many correct solutions, some candidates showed that the helicopter was south or north of the origin, rather than specifically north. Interestingly, some candidates who could not answer part (c) were able to write down the position vector at the specified time. A few candidates found the correct position vector but did not conclude that the helicopter was due north.

When answering part (d)(ii), many of the candidates found the correct velocity, but some did not find the speed as requested.

Q26.

Many candidates made good attempts at parts (a) and (b). The majority of errors were due to errors made manipulating the vectors. Some candidates did write down correct equations or expressions to begin their solutions, but made errors in manipulating their expressions. In part (b), a few candidates included a $75\mathbf{i}$ term.

Several candidates simply did not attempt part (c) of the question. Of those that did, some tried to find a time using expressions for the position of the particle rather than its velocity, while other candidates tried to use the equation $v^2 = u^2 + 2as$.

Q27.

While candidates realised that part (a) required differentiation, a significant number were not able to carry this out accurately. Part (b) proved more successful, but only a small minority of candidates were able to attempt part (c), the simplest solution involving maximisation of the appropriate trigonometrical function.

Q28.

Parts (a), (b) and (c) of this question were often done very well. In part (a) candidates occasionally gave the answers "5" or "5j", but the vast majority of the answers were correct. In part (b), many correctly stated the velocity, but some candidates then performed incorrect simplifications. Part (c) was also done well by those candidates who had obtained the velocity vector.

Part (d) proved to be more challenging. Some candidates realised that they needed to find the position vector and hence the bearing. These candidates usually made good progress,

although some stopped when they had the position vector and others gave the answer as 52.3° . Some candidates tried to work with velocities.

Q29.

Most candidates completed this question successfully. A few used $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ in part (b) and obtained $\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 6t^2)\mathbf{j}$, rather than the given answer. After they had done the integration of $\mathbf{a}$, a number of candidates 'invented' the $6\mathbf{i} + 30\mathbf{j}$ terms required.

Q30.

Part (a) was done well, but in part (b) many stopped after finding the velocity at the given time, instead of continuing to find the speed by evaluating the magnitude of the velocity vector.

Q31.

In part (a), many of the candidates found the force correctly, but some did not go on to give the magnitude. A few candidates did not substitute $t = 0$. In part (b), the main problem was with the integration of the trigonometric terms. A particular problem was with the signs of the functions. Most candidates did attempt to find the constants of integration. A small number of candidates tried to use $\mathbf{v} = \mathbf{u} + \mathbf{a}t$.

Q32.

This was found to be the most difficult question on the paper. While the most able were able to score full marks, others showed confusion between vectors and scalars, or simply showed misunderstanding of vector quantities. Many found the vector $\vec{AB}$ in part (a) but could not then proceed correctly, with ratios of vectors often leading to vector expressions for the time. The most successful methods included the application of the equations of motion with constant acceleration, and the use of diagrams with explained steps in motion. Part (b) often began with the assumption of a zero initial velocity, some only considered the $\mathbf{j}$ component of the motion, and many failed to add the position vector of the particle at B. Part (c) often began well with the vector $\vec{AC}$ found, but with no attempt to find its magnitude to give the distance requested.

Q33.

There were very many good solutions to this question and a good number of candidates scored full marks. Some candidates did however make errors. In part (a) there were some cases where the candidates made some minor errors when differentiating one or both of the components. In part (b) a few candidates had difficulties substituting $t = 1/3$ correctly, but the biggest problem in this part of the question was the description of the direction. While a number did state that the direction was south, some stated that it was north, even though they had calculated the velocity correctly. Others did not use either north or south, but gave answers that implied that $\mathbf{j}$ was a vertical unit vector.

Parts (c) and (d) were generally done well, but a few candidates experienced difficulties with the differentiation and very occasionally with the substitution of $t = 4$.

Q34.

Candidates found this question quite demanding. Candidates were most successful with part (a) and for some this was the only place where they gained marks. In part (b), many candidates omitted the initial position of the yacht, and a number tried to integrate the initial velocity. In part (c)(i), a poor answer for the position vector had an impact. However, some candidates who had good expressions simply showed that the i component of the position was zero and not that the yacht was due north of the origin. In part (c)(ii), quite a few candidates found the velocity correctly, but did not go on to find the speed.

Q35.

There were many good responses to this question. Generally part (a) was done very well. Part (b) was most demanding, but many candidates still did well on this part. One error was to work with a position vector instead of a velocity vector. In part (c), a few candidates gave the velocity as their final answer instead of the speed.



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